

Missing Values and Outlier Detection

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Introduction

This document describes the focused analysis of missing values and outliers applied to the Formula 1 reconciled database.

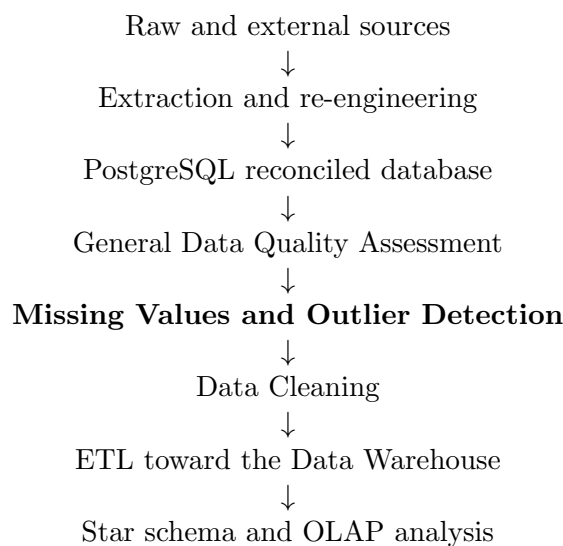
This phase is separated from the General Data Quality Assessment. Some data quality problems require a more specific interpretation. This is especially true for missing values and numerical outliers. In Formula 1 data, a missing value is not always an error. Similarly, an extreme lap time or speed value is not always wrong. It may be caused by pit stops, Safety Car periods, deleted laps, inaccurate laps, rain, crashes, red flags, or other racing situations.

For this reason, this focused analysis is designed as a separate step after the General DQA and before the final Data Cleaning phase.

The goal is not to clean the data automatically. The goal is to classify missing values, flag suspicious numerical values, and produce evidence for later cleaning or analytical filtering decisions.

Position in the Project Pipeline

The focused missing values and outlier detection phase is placed after the General DQA.



Scope of the Focused Analysis

The focused analysis is applied only to the two most important and most complex tables:

- `lap`
- `result`

These tables are selected because they contain the main analytical data used in the project.

Missing Values

The goal of the focused missing values analysis is not only to count null values. The goal is to interpret why a value is missing and how much this missing value affects the analytical use of the row.

Missing Value Classes

The script uses two main missing value classes.

Class	Meaning	Example
EXPLAINED_NULL	The value is missing, but the missingness can be explained by a clear Formula 1 domain rule or by another observed attribute in the row.	<code>q2_ms</code> is missing because the driver was eliminated in Q1; <code>pit_in_time_ms</code> is missing because the lap is not an in-lap.
SUSPICIOUS_NULL	The value is missing and the available domain rules do not provide a clear explanation.	<code>lap_time_ms</code> is missing even though the lap is not a pit-in lap, not a pit-out lap, not deleted, and marked as accurate.

Missing Information Area

In addition to the missing value class, the script assigns a second field called `missing_information_area`.

The goal of this field is to describe which type of analytical information is missing from the row.

This choice is more useful than a generic impact level because the same missing value can affect different analyses in different ways.

For example, a lap with a missing speed value may still be usable for lap time analysis, but it cannot be used for speed-based analysis.

Similarly, a lap with a missing compound may still contain a lap time, but it cannot be used for tyre-based or fair contextual comparison.

Therefore, the script does not directly decide which dashboard or analysis must exclude the row.

Instead, it records the type of missing information. The later visualization phase can decide how to use or filter the row depending on the analytical goal.

Missing Information Area	Meaning
NONE	The missing value does not remove useful analytical information. This is used for expected nulls.
LAP_TIME_INFORMATION	The missing value affects the main lap time information. This is used when <code>lap_time_ms</code> is missing.
SECTOR_INFORMATION	The missing value affects sector-based performance information. This is used when one or more sector times are missing.
SPEED_INFORMATION	The missing value affects speed-based information. This is used when one or more speed attributes are missing.
TYRE_INFORMATION	The missing value affects tyre-related information. This is used when attributes such as <code>compound</code> or <code>tyre_life</code> are missing.

Missing Information Area	Meaning
WEATHER_INFORMATION	The missing value affects weather context information. This can be used when a lap cannot be associated with reliable weather information.
TRACK_STATUS_INFORMATION	The missing value affects track condition information. This can be used when the track status context is missing or not interpretable.
PIT_INFORMATION	The missing value affects pit-related information. This can be used when pit information is required to interpret special laps.

For the `result` table, the following information areas are used:

Missing Information Area	Meaning
NONE	The missing value does not remove useful analytical information in the current session context. This is used for expected nulls, such as qualifying times in Race sessions or race-related attributes in Qualifying sessions.
CLASSIFICATION_INFORMATION	The missing value affects the classification information of the driver in the session. This is used when <code>position</code> is missing, or when <code>classified_position</code> is missing in a Race session. This area is considered critical because classification information is necessary to interpret the final order or result of the driver.
QUALIFYING_INFORMATION	The missing value affects qualifying performance information. This is used when <code>q1_ms</code> , <code>q2_ms</code> , or <code>q3_ms</code> are missing in a context where they should normally be present according to the qualifying progression.
RACE_CONTEXT_INFORMATION	The missing value affects additional race context information. This is used for missing race-related attributes such as <code>grid_position</code> , <code>status</code> , <code>points</code> , <code>laps</code> , or <code>time_ms</code> . These attributes may be useful for specific analyses, but they are not always essential for the main lap performance analysis of the project.

The final interpretation of a missing value is obtained by combining the two fields:

Column	Missing Class	Missing Information Area
<code>q3_ms</code>	EXPLAINED_NULL	NONE
<code>sector1_time_ms</code>	EXPLAINED_NULL	SECTOR_INFORMATION
<code>compound</code>	SUSPICIOUS_NULL	TYRE_INFORMATION
<code>lap_time_ms</code>	SUSPICIOUS_NULL	LAP_TIME_INFORMATION

This approach keeps the classification simple but more useful for the later analytical steps.

The field `missing_class` explains why the value is null. The field `missing_information_area` explains what kind of information is missing from the row.

For example, if a later visualization requires tyre information, it can exclude rows where `missing_information_area` contains `TYRE_INFORMATION`. If a visualization only compares basic lap times, it may still use rows where speed or sector information is missing, as long as the lap time itself is present.

Missing Values on the Result Table

The **result** table contains one row for one driver in one session. It includes both qualifying-related attributes and race-related attributes.

Category	Attributes
Qualifying Attributes	q1_ms, q2_ms, q3_ms
Race-related Attributes	classified_position, status, points, laps, time_ms

Because the meaning of these attributes depends on the session type, missing values in this table are analyzed differently for Qualifying and Race sessions.

The script uses the **session_type** associated with each **session_id** to decide which rules must be applied.

Race Sessions

For Race sessions, qualifying time attributes are not applicable.

Therefore, missing values in **q1_ms**, **q2_ms**, and **q3_ms** are classified as explained nulls with no missing analytical information.

Missing Attribute	Missing Class	Missing Information Area
q1_ms	EXPLAINED_NULL	NONE
q2_ms	EXPLAINED_NULL	NONE
q3_ms	EXPLAINED_NULL	NONE

These values are explained by the session type: a Race session does not contain Q1, Q2, or Q3 segment times. Therefore, no useful race-result information is missing.

By contrast, race-related attributes such as **classified_position**, **status**, **points**, **laps**, and **time_ms** are expected to be present in Race sessions. If one of these values is missing, the missing value is classified as suspicious because race-result information is incomplete.

Missing Attribute	Missing Class	Missing Information Area
position	SUSPICIOUS_NULL	CLASSIFICATION_INFORMATION
classified_position	SUSPICIOUS_NULL	CLASSIFICATION_INFORMATION
grid_position	SUSPICIOUS_NULL	RACE_CONTEXT_INFORMATION
status	SUSPICIOUS_NULL	RACE_CONTEXT_INFORMATION
points	SUSPICIOUS_NULL	RACE_CONTEXT_INFORMATION
laps	SUSPICIOUS_NULL	RACE_CONTEXT_INFORMATION
time_ms	SUSPICIOUS_NULL	RACE_CONTEXT_INFORMATION

Qualifying Sessions

In Qualifying sessions, **position** is fundamental because it is used to interpret the qualifying progression. Without **position**, the script cannot reliably decide whether a missing **q2_ms** or **q3_ms** is expected or suspicious.

Missing Attribute	Missing Class	Missing Information Area
position	SUSPICIOUS_NULL	CLASSIFICATION_INFORMATION

For Qualifying sessions, the attributes `q1_ms`, `q2_ms`, and `q3_ms` must be interpreted according to the qualifying progression.

Not all drivers are expected to have Q2 or Q3 times. Drivers eliminated in Q1 should normally have no Q2 or Q3 time. Drivers eliminated in Q2 should normally have no Q3 time.

For this reason, missing qualifying times can be either explained or suspicious, depending on the final qualifying position.

Condition	Missing Attribute	Missing Class	Missing Information Area
<code>position > 15</code>	<code>q2_ms</code>	EXPLAINED_NULL	NONE
<code>position > 15</code>	<code>q3_ms</code>	EXPLAINED_NULL	NONE
<code>11 <= position <= 15</code>	<code>q3_ms</code>	EXPLAINED_NULL	NONE

These missing values are not considered data quality errors because they are explained by the structure of the qualifying session. In these cases, no useful qualifying information is missing, because the driver did not reach the corresponding qualifying phase.

However, if a qualifying time is missing even though the driver's position indicates that the driver should normally have reached that phase, the missing value is classified as suspicious and the missing information area is `QUALIFYING_INFORMATION`.

Condition	Missing Attribute	Missing Class	Missing Information Area
Qualifying session	<code>q1_ms</code>	SUSPICIOUS_NULL	QUALIFYING_INFORMATION
<code>position <= 15</code>	<code>q2_ms</code>	SUSPICIOUS_NULL	QUALIFYING_INFORMATION
<code>position <= 10</code>	<code>q3_ms</code>	SUSPICIOUS_NULL	QUALIFYING_INFORMATION

In these cases, qualifying information is incomplete for that row. For example, if a driver is classified in the top 10 but `q3_ms` is missing, the script flags the value as `SUSPICIOUS_NULL` with `QUALIFYING_INFORMATION`.

The row is not corrected automatically, because special cases may exist, such as drivers who did not set a valid time, deleted laps, penalties, or session anomalies. The script only records that qualifying information is missing and that the case requires review.

For Qualifying sessions, race-specific attributes such as `points`, `laps`, `classified_position`, `status`, and `time_ms` are not central to the qualifying performance analysis.

If these attributes are missing in a Qualifying session, the missing values are treated as explained by the session type and no useful qualifying information is considered missing.

Missing Attribute	Missing Class	Missing Information Area
<code>classified_position</code>	EXPLAINED_NULL	NONE
<code>status</code>	EXPLAINED_NULL	NONE
<code>points</code>	EXPLAINED_NULL	NONE
<code>laps</code>	EXPLAINED_NULL	NONE
<code>time_ms</code>	EXPLAINED_NULL	NONE
<code>grid_position</code>	EXPLAINED_NULL	NONE

This rule keeps the analysis simple and avoids flagging expected null values as quality problems.

Missing Values on the Lap Table

The `lap` table is the most complex table of the reconciled database. Each row represents one lap performed by one driver in one session.

This table contains several analytical attributes that are important for lap performance analysis:

Timing attributes	Speed attributes	Tyre / pit attributes
<code>lap_time_ms</code>	<code>speed_i1</code>	<code>compound</code>
<code>sector1_time_ms</code>	<code>speed_i2</code>	<code>tyre_life</code>
<code>sector2_time_ms</code>	<code>speed_fl</code>	<code>pit_in_time_ms</code>
<code>sector3_time_ms</code>	<code>speed_st</code>	<code>pit_out_time_ms</code>

Pit-Related Missing Values

The attributes `pit_in_time_ms` and `pit_out_time_ms` represent session timestamps.

If `pit_in_time_ms` is present, the lap is an in-lap. If `pit_out_time_ms` is present, the lap is an out-lap.

These attributes are normally null in clean laps where the driver does not enter or exit the pit lane. Therefore, missing values in these two columns are usually expected and do not represent data quality problems.

Missing Attribute	Condition	Missing Class	Missing Information Area
<code>pit_in_time_ms</code>	The lap is not an in-lap.	EXPLAINED_NULL	NONE
<code>pit_out_time_ms</code>	The lap is not an out-lap.	EXPLAINED_NULL	NONE
<code>deleted_reason</code>	<code>deleted = false</code>	EXPLAINED_NULL	NONE

Core Lap Performance Missing Values

The attribute `lap_time_ms` is the core measure of lap performance. If this value is missing, the row lacks the main lap time information.

Condition	Missing Attribute	Missing Class	Missing Information Area
No deletion or inaccuracy explanation is available	<code>lap_time_ms</code>	SUSPICIOUS_NULL	LAP_TIME_INFORMATION

Sector Time Missing Values

Sector times are important for sector-based performance analysis. However, missing sector times may occur in special laps.

The script applies a conservative interpretation. A missing sector time is classified as `EXPLAINED_NULL` only when there is a direct domain explanation.

In this project, the only direct pit-related explanation used for sector times is the presence of `pit_in_time_ms` for a missing `sector3_time_ms`. This is because `pit_in_time_ms` means that the driver entered the pit lane during that lap, and the final part of the lap may be affected by pit entry or by the way the source records in-laps.

By contrast, `pit_out_time_ms` is not used as an automatic explanation for a missing `sector1_time_ms`. Even if the driver exits the pit lane during the lap, the first sector should not be considered missing by default only because of the pit exit information.

Condition	Missing Attribute	Missing Class	Missing Information Area
<code>pit_in_time_ms</code> is present	<code>sector3_time_ms</code>	EXPLAINED_NULL	SECTOR_INFORMATION
No direct sector-specific explanation is available	Any missing sector time	SUSPICIOUS_NULL	SECTOR_INFORMATION

A missing sector time always means that sector information is incomplete. The difference between `EXPLAINED_NULL` and `SUSPICIOUS_NULL` only depends on whether the script can find a direct and specific explanation.

Speed Missing Values

Speed attributes such as `speed_i1`, `speed_i2`, `speed_fl`, and `speed_st` are useful for speed-based analysis.

A missing speed value does not necessarily make the whole lap unusable, but it means that speed information is incomplete.

The script applies a conservative domain interpretation. Pit-related information is not used to explain all speed missing values. It is used only for the speed attributes that can reasonably be affected by pit entry, pit exit, or the start/end area of the lap.

In this project, when `pit_in_time_ms` or `pit_out_time_ms` is present, missing values in `speed_i1`, `speed_fl`, and `speed_st` can be considered explained. By contrast, missing `speed_i2` is not automatically explained by pit information, because it is not directly associated with the pit entry or pit exit interpretation used in this simplified rule set.

Condition	Missing Attribute	Missing Class	Missing Information Area
<code>pit_in_time_ms</code> is present	<code>speed_i1</code> , <code>speed_fl</code> , or <code>speed_st</code>	EXPLAINED_NULL	SPEED_INFORMATION
<code>pit_out_time_ms</code> is present	<code>speed_i1</code> , <code>speed_fl</code> , or <code>speed_st</code>	EXPLAINED_NULL	SPEED_INFORMATION
Only pit information is present	<code>speed_i2</code>	SUSPICIOUS_NULL	SPEED_INFORMATION
No pit-related explanation, not deleted, and accurate lap	Any missing speed attribute	SUSPICIOUS_NULL	SPEED_INFORMATION

This rule avoids over-explaining speed missing values. The presence of an in-lap or out-lap is used as a possible explanation only for selected speed attributes. Missing `speed_i2` is treated more cautiously and is classified as suspicious unless another explanation, such as `deleted = true` or `is_accurate = false`, is available.

Tyre and Compound Missing Values

The attribute `compound` is essential in this project because lap performance cannot be interpreted fairly without knowing the tyre compound.

Two laps with the same `lap_time_ms` may have very different meanings if they were set on different compounds. For this reason, a missing `compound` means that tyre information is incomplete.

Condition	Missing Attribute	Missing Class	Missing Information Area
Any lap	compound	SUSPICIOUS_NULL	TYRE_INFORMATION
Any lap	tyre_life	SUSPICIOUS_NULL	TYRE_INFORMATION

The value `tyre_life` is important for tyre degradation and race pace comparison. If `tyre_life` is missing, the row may still contain lap time and compound information, but tyre degradation information is incomplete.

Track Status and Weather Context Missing Values

Track status and weather information are important contextual elements for fair lap comparison.

In the current reconciled `lap` table, `track_status` is already available as a lap-level attribute. If this value is missing or not interpretable, the row lacks track condition information.

Weather information is stored in a separate `weather` table. The focused missing analysis detects source-level nulls; the warehouse ETL subsequently aligns contextual observations to each lap. Within the same `session_id`, both sides are ordered by the session timeline and the lap timestamp is `lap_start_time_ms`, not `lap_time_ms`. The canonical association first selects the latest observation whose `time_ms` is not later than the lap start. If none exists, the nearest observation in that same session is used as a controlled fallback. No association may cross a session boundary; a failed association activates `has_weather_information_issue` or `has_track_status_information_issue` as appropriate.

Condition	Missing Attribute	Missing Class	Missing Information Area
Any lap	<code>track_status</code>	SUSPICIOUS_NULL	TRACK_STATUS_INFORMATION
No same-session weather observation can be associated	Derived lap weather context	SUSPICIOUS_NULL	WEATHER_INFORMATION

This separation keeps source missingness detection distinct from contextual enrichment and from the cleaning decisions that consume the resulting quality flags.

Outlier Detection

Outlier detection is used to identify numerical values that are very different from the majority of values.

In this project, outliers are not automatically considered errors. Formula 1 data naturally contains extreme values caused by race context.

For example:

- a slow sector may be caused by yellow flags;
- a low speed value may be caused by traffic or Safety Car;
- a very high lap time may be caused by rain, red flags, or an incomplete lap.

Therefore, outlier detection is used only to flag suspicious values for review.

Methods Used

The script uses two simple and interpretable univariate methods:

- IQR method
- Modified Z-score

IQR Method

The Interquartile Range method is based on the first quartile, the third quartile, and the interquartile range.

$$IQR = Q3 - Q1$$

The lower and upper bounds are defined as:

$$LowerBound = Q1 - k \cdot IQR$$

$$UpperBound = Q3 + k \cdot IQR$$

A value outside this interval is flagged as an outlier.

In this project, the IQR method is useful because it does not require a normal distribution assumption.

Modified Z-score

The Modified Z-score is based on the median and the Median Absolute Deviation.

$$M_i = \frac{0.6745(x_i - median(x))}{MAD}$$

where:

$$MAD = median(|x_i - median(x)|)$$

A value is flagged as suspicious when:

$$|M_i| > 3.5$$

This method is useful for Formula 1 data because lap times and speed values may be skewed by pit stops, special laps, or race interruptions.

Consensus Strategy

No single outlier detection method is perfect. For this reason, the script uses a simple consensus strategy.

Each method can flag a value. The number of methods that flag the same value is counted.

Consensus Score	Meaning	Action
0	Not flagged by any method.	Normal value.
1	Flagged by one method.	Weak anomaly. Review only if relevant for the analysis.

Consensus Score	Meaning	Action
2	Flagged by both IQR and Modified Z-score.	Stronger suspicious value. Investigate before using the row in sensitive analysis.

The detection script does not remove consensus outliers: it only records flags and the consensus score. The downstream cleaning phase applies the decision policy. A `lap_time_ms` value with consensus score 2 causes `DROP_ROW`; sector or speed outliers preserve the lap and activate the corresponding quality flag.

Outlier Detection on the Lap Table

The main outlier detection activity is applied to the `lap` table.

The selected numerical attributes are:

Timing Attributes	Timing Attributes	Speed Attributes
<code>lap_time_ms</code>	<code>sector3_time_ms</code>	<code>speed_i1</code>
<code>sector1_time_ms</code>		<code>speed_i2</code>
<code>sector2_time_ms</code>		<code>speed_f1</code>
		<code>speed_st</code>

Outliers are detected within each `session_id`, but the analysis is not applied to all laps of the session.

This is important because lap times and speed values are strongly affected by the racing context. A slow lap is not necessarily a data error. It may be caused by rain, Safety Car, Virtual Safety Car or other racing situations.

For this reason, the script applies outlier detection only to laps that are suitable for performance comparison.

Clean Lap Selection

Before applying outlier detection, the script selects only clean candidate laps.

A lap is considered a clean candidate for outlier detection when the following conditions are satisfied:

Clean Candidate Conditions for Outlier Detection
<code>lap_time_ms</code> is present
<code>compound</code> is present
<code>pit_in_time_ms</code> is null
<code>pit_out_time_ms</code> is null
<code>track_status</code> = 1 (AllClear)
The corresponding session is classified as dry

The goal of this filter is not to clean the data. The goal is to avoid detecting obvious race-context effects as statistical outliers.

For example, a lap under Safety Car would probably be detected as an outlier because it is much slower than normal laps. However, this does not mean that the value is wrong. It means that the lap belongs to a different racing context.

Dry Session Selection

Weather information is stored in the `weather` table as time-varying observations associated with each session.

The outlier detector deliberately uses a conservative session-level rule rather than repeating the lap-level ETL association described above:

A session is considered dry only if all weather observations for that session have `rainfall = false`.

Therefore, if at least one weather observation in a session has `rainfall = true`, the whole session is excluded from lap outlier detection.

This choice may exclude some usable dry laps from mixed sessions, but it avoids incorrectly comparing dry laps and wet laps in the same statistical group.

Track Status Filter

The `lap` table already contains a `track_status` attribute. In this project, the outlier detection script keeps only laps with:

`track_status = 1`

where status code 1 means `AllClear`.

Laps with Safety Car, Virtual Safety Car, yellow flag, red flag, or other abnormal track conditions are excluded from the outlier detection sample.

This is necessary because these laps are expected to have abnormal lap times, sector times, or speed values for domain reasons.

Session-Level Outlier Detection

After applying the clean lap, dry session, and track status filters, the script groups the remaining laps by `session_id`.

For each selected metric, the script applies the IQR method and the Modified Z-score inside each session group.

The comparison is performed within the same session because lap times and speeds are strongly dependent on the circuit and session context. Comparing all laps across the entire dataset would be misleading. For example, Monaco and Monza naturally have very different lap times and speed profiles.

The script applies the outlier methods only when a session contains a sufficient number of valid observations for the selected metric. If too few values are available, the check is skipped for that metric and session.

The output includes:

Recorded Information for Each Outlier Candidate

Row identifier

Session identifier

Metric name

Observed value

IQR method flag

Modified Z-score flag

Consensus score

Interpretation label

Interpretation

A flagged lap value does not automatically mean that the value is wrong.

Even after filtering, an outlier may still be caused by domain reasons such as traffic, driver mistakes, tyre degradation, unusual driving strategy, or source-specific recording behavior.

For this reason, the detector only flags outlier candidates. The separate Data Cleaning phase applies the consensus-based policy described above.

The focused script therefore produces evidence, not automatic corrections.

Output Files

The focused scripts produce four main output files.

Output File	Purpose
<code>focused_missing_summary.csv</code>	Summary of missing values by table and column. It reports the number and percentage of missing values and the main interpretation.
<code>focused_missing_row_flags.csv</code>	Row-level missing value flags. It reports the table, row identifier, column name, missing class, explanation, and severity.
<code>lap_outlier_flags.csv</code>	Row-level outlier flags for lap-level metrics. It reports the metric, value, IQR flag, Modified Z-score flag, and consensus score.
<code>outlier_summary.csv</code>	Aggregated summary of detected outliers by table and metric.